

BE/APh161: Physical Biology of the Cell

Homework 7

Due Date: Weds., March 4, 2026

“How can the events in *space and time* which take place within the spatial boundary of a living organism be accounted for by physics and chemistry?”
- Erwin Schrödinger **What is Life?**

NOTE: As I mentioned in class, this homework is an adaptation from my book and hence has a number of rough edges. I basically edited the narrative that we wrote and turned it into a sort of progression of “show this” and “show that” but the logic might not be perfectly clean as a problem set rather than a narrative. As I also said in class, the idea here is precisely for you to construct a narrative. That is, I want you to tell the story of sugar transport mathematically (and with words) such that you are sure to explain how the net entropy increases (or alternatively, how the net free energy goes down), what I mean by AND gate, how to compute the entropy in lattice models and what it means, etc.

1. The Failure of Equilibrium Fidelity.

In class, I worked out in words that for a simple two amino-acid view of protein translation, that the error rate is given by the ratio of the K_d s of the wrong and the right tRNAs. In this problem, flesh out the entire argument given in class, and generalize beyond the case done in class to the case in which the concentration of the wrong and right tRNAs is different. That is, find an expression for the error rate in this case. Given the real biological situation, briefly explain why the number of wrong tRNAs is higher than the number of right tRNAs. Make sure you put in some approximate (but well justified) numbers. Make a plot of the translational fidelity as a function of the difference in energy between binding the right and the wrong tRNA. Explain why this model *fails* as a picture of translational fidelity. Finish this problem by giving a one-paragraph summary in your own words (not AI generated) of kinetic proofreading. What did Ninio and Hopfield accomplish in their invention of the concept of kinetic proofreading? How does it jibe with my bigger theme in this part of the course about the “secret of life?”

2. Taking Sugars Up Their Gradient

In class, I worked out the dynamics of membrane proteins that act as AND gates resulting in moving sugars up their concentration gradient. In Figure 1, we consider the action of a symporter, a membrane secondary transport protein, that uses the free energy advantage of letting ions go down their gradient to simultaneously pump sugar molecules up their gradient. This simple process has all the key ingredients to help us understand much about nonequilibrium steady states more generally.

As we did in class, the language we will use to mathematicize the action of the symporter is that of entropy. For the situation shown as “BEFORE” in Figure 1, the total entropy of that little world could be increased by allowing both the ions and sugars to move down their respective gradients. But one of the most powerful aspects of many protein machines is that they are engines of constraint. They don’t work unless several reactions happen simultaneously. Here, we make a caricature of that process by thinking of the symporter as an AND gate since it requires both an ion and a sugar to be bound to induce the conformational change that permits transport. Of course, all allosteric molecules have leakiness which means that the backward reaction must happen as well, but for the purposes of this immediate discussion we assume that the reaction is one way.

(A) Following in the footsteps of Gibbs, we adopt entropy maximization as our way of thinking about the driving force. In particular, during one reaction of the symporter, show that we can write the entropy change as

$$\Delta S = \left(\frac{\partial S}{\partial N_{sugar}^{(in)}} \right) dN_{sugar}^{(in)} + \left(\frac{\partial S}{\partial N_{sugar}^{(out)}} \right) dN_{sugar}^{(out)} + \left(\frac{\partial S}{\partial N_{ion}^{(in)}} \right) dN_{ion}^{(in)} + \left(\frac{\partial S}{\partial N_{ion}^{(out)}} \right) dN_{ion}^{(out)}. \quad (1)$$

As we have already remarked, the mechanistic feature that makes these biological reactions work is coupling. That is, when one thing changes, so too must another which is coupled. For example, in the case of the symporter, the magic of the protein is that it serves as an AND gate and only opens when charged with both an ion and a sugar. At that point, when the ion and sugar are transported across the membrane we know that $dN_{sugar}^{(out)} = -dN_{sugar}^{(in)}$. Specifically, when $N_{sugar}^{(in)}$ goes up by one, then $N_{sugar}^{(out)}$ goes down by one. Similarly, when $N_{sugar}^{(in)}$ goes up by one, so too does $N_{ion}^{(in)}$. Finally, when $N_{sugar}^{(in)}$ goes up by one, $N_{ion}^{(out)}$ goes down by one.

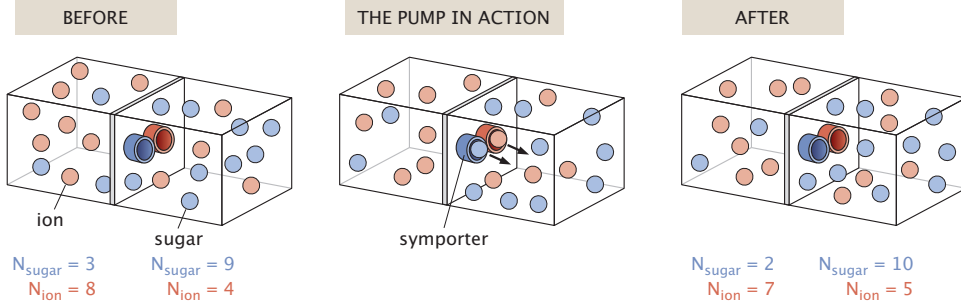


Figure 1: Pumping sugar up its gradient with a membrane protein known as a symporter. As seen in the “BEFORE” image, there are more sugars in the right side than the left, while there are more ions to the left than to the right. The symporter protein serves as an AND gate, only opening when both the ion and sugar are bound to it, resulting in a reaction which increases the total entropy, but attended by the happy circumstance of pumping sugars up their concentration gradient. The number of sugars and ions on both sides both before and after the transport event are shown for clarity.

(B) Show that the AND character of the symporter means that

$$dN_{\text{sugar}}^{(in)} = dN_{\text{ion}}^{(in)} = -dN_{\text{sugar}}^{(out)} = -dN_{\text{ion}}^{(out)}. \quad (2)$$

Essentially, by keeping track of the number of sugars inside the cell, we can keep track of the rest of the molecular populations because of this constraint. We have a one-parameter problem in which everything depends upon $N_{\text{sugar}}^{(in)}$.

(C) Show that we can rewrite the expression above for the change in entropy as

$$\Delta S = \left[\left(\frac{\partial S}{\partial N_{\text{sugar}}^{(in)}} \right) - \left(\frac{\partial S}{\partial N_{\text{sugar}}^{(out)}} \right) + \left(\frac{\partial S}{\partial N_{\text{ion}}^{(in)}} \right) - \left(\frac{\partial S}{\partial N_{\text{ion}}^{(out)}} \right) \right] dN_{\text{sugar}}^{(in)}. \quad (3)$$

But now we can go further since we recognize that the chemical potential is defined precisely as $\mu = (\partial F / \partial N)_{V,T}$, where F is the free energy $F = U - TS$. Note that when we write $(\partial F / \partial N)_{V,T}$, the subscript V,T simply means that we are holding V and T constant while varying N . To be explicit about the functional form of the chemical potential, in our idealized

solution model, we assume that the free energy $F = U - TS$ can be written as a constant term $U = \mu_0$ and then an entropy term.

(D) Show that for a sugar molecule in an ideal solution the chemical potential can be written as

$$\mu_{sugar}^{(in)} = \mu_{sugar}^{(0)} + k_B T \ln \frac{(N_{sugar}^{(in)}/V_{in})}{c_0} \quad (4)$$

for the sugars in the cell, with a similar expression for the sugars outside of the cell which we write as

$$\mu_{sugar}^{(out)} = \mu_{sugar}^{(0)} + k_B T \ln \frac{(N_{sugar}^{(out)}/V_{out})}{c_0}. \quad (5)$$

We have corresponding expressions for the ions which we write as

$$\mu_{ion}^{(in)} = \mu_{ion}^{(0)} + k_B T \ln \frac{(N_{ion}^{(in)}/V_{in})}{c_0} \quad (6)$$

and

$$\mu_{ion}^{(out)} = \mu_{ion}^{(0)} + k_B T \ln \frac{(N_{ion}^{(out)}/V_{out})}{c_0}. \quad (7)$$

To do this part of the problem, make a lattice model of solution like I did in class and show how the entropy of each reservoir changes when you add or remove a single molecule and show how that gives rise to the expressions above. Make sure you carefully explain what c_0 is and why it must be there on dimensional grounds. Given that we took a lattice model and asserted a volume of 1 nm^3 for each lattice site, what is c_0 in M units?

(D) As a result of these chemical potential definitions, show that we can rewrite the net free energy change during one symport transport event as

$$\Delta F = \left[\mu_{sugar}^{(in)} - \mu_{sugar}^{(out)} + \mu_{ion}^{(in)} - \mu_{ion}^{(out)} \right] dN_{sugar}^{(in)} \leq 0. \quad (8)$$

The reason we write this as an inequality is that the transport process takes the system downhill in free energy. Indeed, that is the power of the variational statement of equilibrium of Gibbs in which he instructs us to determine the terminal privileged state of a system by minimizing its free energy. Hence,

when $\left[\mu_{sugar}^{(in)} - \mu_{sugar}^{(out)} + \mu_{ion}^{(in)} - \mu_{ion}^{(out)} \right] \neq 0$, this measure of the degree of disequilibrium quantifies the driving force towards equilibrium.

(E) In light of the definitions of the chemical potentials given above, show that we can rewrite our expression for the driving force as

$$\text{driving force} = k_B T \ln \left(\frac{c_{sugar}^{(in)} c_{ion}^{(in)}}{c_{sugar}^{(out)} c_{ion}^{(out)}} \right), \quad (9)$$

implying that in equilibrium, we have

$$\frac{c_{sugar}^{(in)} c_{ion}^{(in)}}{c_{sugar}^{(out)} c_{ion}^{(out)}} = 1, \quad (10)$$

since $\ln 1 = 0$, which implies the vanishing of the driving force. This recasting of the equilibrium condition will prove useful when we examine this same problem from the point of view of chemical kinetics.

We are now poised to see the consequences of this simple one-parameter model. The entire mathematical description of this problem can be evaluated strictly in terms of the number of sugar molecules in the cell because the change in the number of all molecular and ionic counts are quantitatively constrained to change with $N_{sugar}^{(in)}$ as we saw in eqn. 2. To determine the dynamics, we adopt the powerful phenomenological dynamics described by the kinetic law

$$\frac{\partial N_{sugar}^{(in)}}{\partial t} = -\Gamma \frac{\partial F}{\partial N_{sugar}^{(in)}}. \quad (11)$$

This phenomenological approach to out of equilibrium dynamics says that the rate of relaxation towards the equilibrium state is proportional to the degree of disequilibrium as measured by the gradient in the free energy. Interestingly, for small departures from equilibrium, this approach yields Fick's law for diffusion which implies a linear relationship between the rate of transport and the degree of disequilibrium as measured by the concentration gradient. Figure 2 shows the driving force of eqn. 8 as a function of $N_{sugar}^{(in)}$. Close to the steady state value of $N_{sugar}^{(in)}/N = 0.75$, the driving force in the vicinity of this equilibrium is linear in the departure from equilibrium in accordance with Fick's law, but then adopts a highly nonlinear profile for larger departures from the steady state.

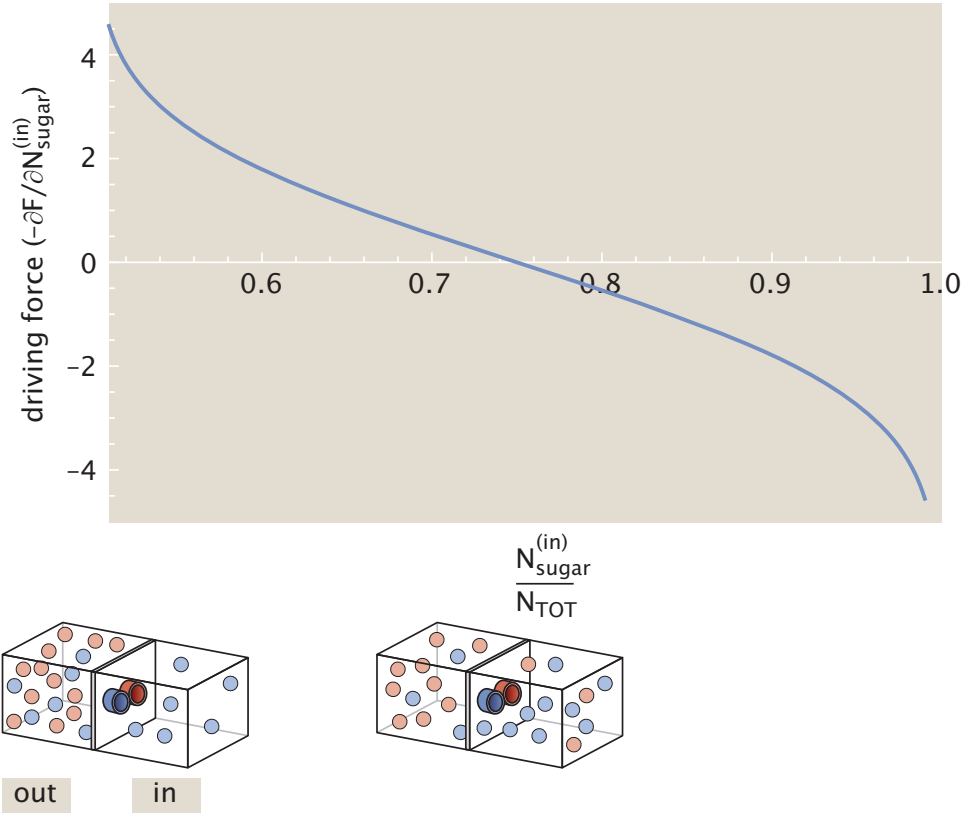


Figure 2: Free-energy driven kinetics of a symporter. The x-axis shows the normalized number of sugar molecules in the cell. The y-axis reports on the driving force as defined in the right side of eqn. 11. We see that the driving force vanishes when 75% of the sugar molecules are in the cell.

To give a sense of the dynamics that results from these free energy relaxation arguments, Figure 3 shows the concentration of the sugar and ions both outside and inside of the cell as a function of time, given an initial condition in which all of the ions are outside of the cell and the number of sugars outside and inside the cell is equal. We see that, as expected, the number of ions on the outside decreases as they accompany the sugars up their concentration gradient. Similarly, the number of sugars outside goes down and the number of sugars on the inside goes up. However, we also see that after a certain time, the system reaches its free energy minimum state and all further transport ceases. Said differently, the biological battery in the form of the difference in ion concentration across the membrane has run out of charge. In reality, as we will discuss below after providing a few more perspectives on the present problem, the battery is constantly recharged as ions are pumped back out of the cell.

It is of great interest to recast this entire discussion of the symporter in the language of rate equations. Indeed, we will permit ourselves to pass back and forth freely between the free energy perspective and the rate equation perspective. One of the advantages of the rate equation perspective is that we will be able to explicitly see how the sugar transport process breaks detailed balance.

(F) Show that we can write a rate equation describing the number of sugars inside the cell as

$$\frac{dc_s^{(\text{in})}}{dt} = k_+ c_s^{(\text{out})} c_i^{(\text{out})} - k_- c_s^{(\text{in})} c_i^{(\text{in})} \quad (12)$$

where we introduce the notation $c_s^{(\text{in})}$ for the concentration of sugars in the cell, $c_i^{(\text{out})}$ for the concentration of ions outside of the cell and so on. We see that this equation explicitly accounts for the *coupling* in the sense that to transport a sugar requires both a sugar and an ion. The equilibrium state corresponds to the state in which $\frac{dc_s^{(\text{in})}}{dt} = 0$ which implies in turn that

$$\frac{c_s^{(\text{in})}(\text{eq}) c_i^{(\text{in})}(\text{eq})}{c_s^{(\text{out})}(\text{eq}) c_i^{(\text{out})}(\text{eq})} = \frac{k_+}{k_-} = 1. \quad (13)$$

What makes molecular machines such as symporters so interesting is that they are driven by a steady, nonequilibrium concentration of ions as shown in Figure 4. Hence, every time that an ion goes down its gradient, an ion is

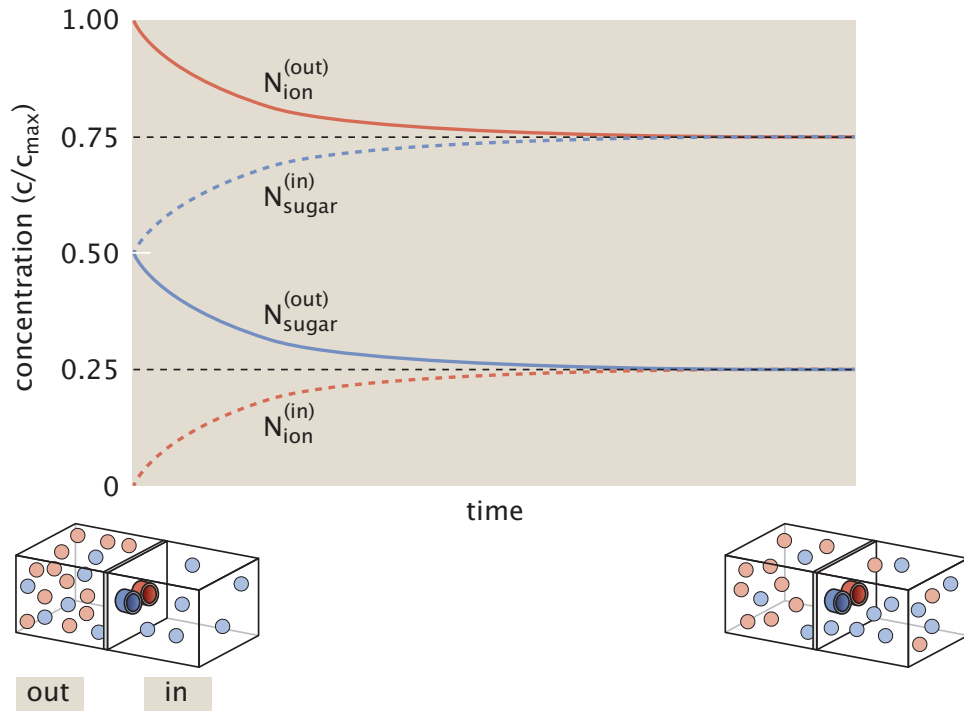


Figure 3: Dynamics of the symporter. Given an initial condition in which all of the ions start on the outside and there is an equal number of sugars on the inside and outside, the system evolves over time until it reaches its free energy minimum at which $3/4$ of the sugars are on the inside and $1/4$ of the sugars are on the outside.

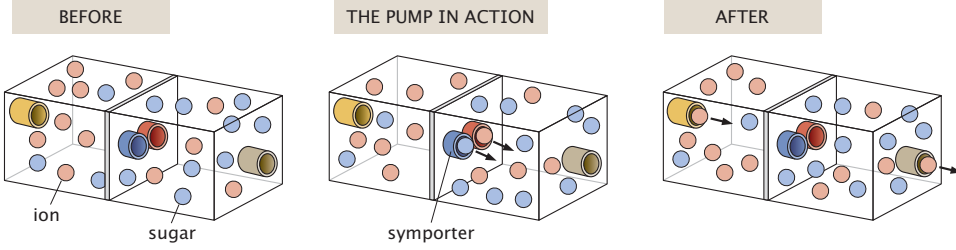


Figure 4: Symporters maintained in a state of readiness. The ion gradient that pays the free energy of the symporter process is maintained in a nonequilibrium steady state through the action of pumps (shown in yellow). For every ion that is used by the symporter, a new one is added on the outside and one is removed from the inside, thus maintaining the ion gradient driving force. This schematic introduces the fiction of the pumps connected to the two boxes, whereas in the cell the way this really happens is that a separate protein pumps hydrogen ions back across the membrane from the inside of the cell to the outside of the cell.

added back to the outside and removed from the inside. At first, this might seem to violate the conservation of energy, but recall that the energy available from the sugar molecules is of order $1000 k_B T$. As a result, one of the ATPs derived from the sugar molecule can be used to do the ion pumping, still leaving behind a great excess of available ATPs to do other things. We illustrate this idea in Figure 5(A) where we can think of the coupled process of sugar and ion transport into the cell, which brings with it a substantial excess of energy as illustrated in the analogy of Figure 5(B).

(G) Given that the ion concentration is held fixed, show that we can rewrite our dynamical equation of eqn. 12 as a coarse-grained equation that no longer makes explicit reference to the ions as

$$\frac{dc_s^{(in)}}{dt} = k_+^{\text{eff}} c_s^{(out)} - k_-^{\text{eff}} c_s^{(in)}, \quad (14)$$

where we define $k_+^{\text{eff}} = c_i^{(out)} k_+$ and $k_-^{\text{eff}} = c_i^{(in)} k_-$.

(H) Now, show that the steady state condition can be written as

$$\frac{c_s^{(in)}}{c_s^{(out)}} = \frac{k_+^{\text{eff}}}{k_-^{\text{eff}}} = \frac{c_i^{(out)}}{c_i^{(in)}} \frac{k_+}{k_-}. \quad (15)$$

(I) Using the definition of the chemical potential, show that we can rewrite concentration in the form

$$c = c_0 e^{(\mu - \mu_0)/k_B T}, \quad (16)$$

yielding the beautiful result that

$$\frac{k_+^{\text{eff}}}{k_-^{\text{eff}}} = \frac{k_+}{k_-} e^{(\mu_i^{(out)} - \mu_i^{(in)})/k_B T}, \quad (17)$$

demonstrating that the ratio of the rate constants is boosted, signaling that detailed balance has been broken. There is a level of arbitrariness to our choice here in the sense that we can now assign the broken detailed balance to the forward rate k_+ resulting in

$$k_+^{\text{eff}} = k_+ e^{(\mu_i^{(out)} - \mu_i^{(in)})/k_B T}. \quad (18)$$

What has emerged from this simple model system is a very clear picture of the way detailed balance is broken and how the magnitude of the departure from detailed balance is related to the degree of disequilibrium as measured by the free energy

$$\Delta F_{\text{drive}} = \mu_i^{(out)} - \mu_i^{(in)}. \quad (19)$$

When the ions are no longer maintained out of equilibrium, $\Delta F_{\text{drive}} = 0$ and the rate constants return to their detailed balance values.

The point of this long case study was to systematically explore the way in which coupling between thermodynamically unfavorable reactions (such as taking molecules up their concentration gradient) can be coupled to thermodynamically favorable reactions (such as hydrogen ions flowing down their concentration gradient). By studying the active pumping process of a symporter, we have not only shown how this coupling works, but were also able to see why a viable approach to thinking about these problems is to use ideas from equilibrium statistical physics such as the free energy and the chemical potential. Further, by casting our analysis in the language of rate equations, we could explicitly see how the nonequilibrium free energy of the ions serves as a biological battery that breaks detailed balance.

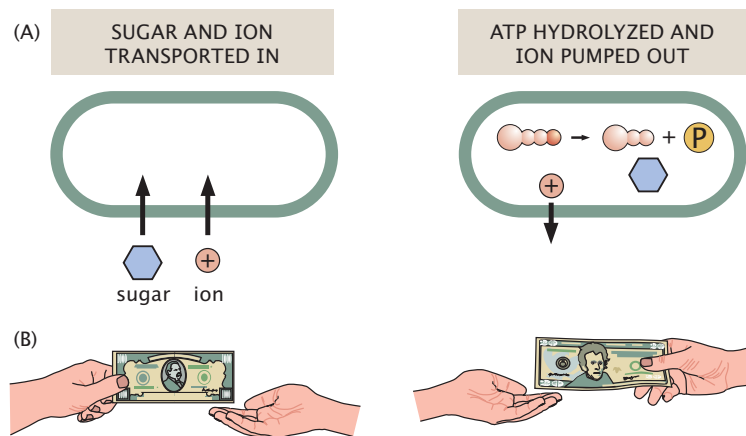


Figure 5: Recharging the biological battery. (A) A single sugar is transported into the cell up its concentration gradient, by surfing on the free energy reducing transport of an ion. An ATP is consumed in order to pump the driving ion back up its gradient. (B) A financial analogy in which every day, one person gives 1000 dollars to a second person and in return, the second person pays the first person 20 dollars.